

Zepto SQL Data Analysis Project

Project Overview

This project performs data exploration, cleaning, and analysis on Zepto product inventory data using SQL.

The goal is to extract business insights related to product pricing, discounts, inventory availability, and revenue potential.

The analysis simulates a real-world e-commerce retail dataset where SQL is used to understand product performance and inventory patterns.

Dataset Description

The dataset contains information about products sold on Zepto, including:

Column	Description
sku_id	Unique product identifier
category	Product category
name	Product name
mrp	Maximum retail price
discountPercent	Discount applied
availableQuantity	Available stock quantity
discountedSellingPrice	Final selling price
weightInGms	Product weight
outOfStock	Product stock status
quantity	Pack quantity

Database Schema:

```
CREATE TABLE zepto (  
  sku_id SERIAL PRIMARY KEY,  
  category VARCHAR(120),
```

```
name VARCHAR(150) NOT NULL,  
mrp NUMERIC(8,2),  
discountPercent NUMERIC(5,2),  
availableQuantity INTEGER,  
discountedSellingPrice NUMERIC(8,2),  
weightInGms INTEGER,  
outOfStock BOOLEAN,  
quantity INTEGER  
);
```

Data exploration

Count of rows

```
select count(*) from zepto;
```

Data Output	
1	count bigint 3732

Sample data

```
SELECT * FROM zepto
```

LIMIT 10;

Data Output

Messages

Notifications

SQL

Showing rows: 1 to 10

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of 1

	sku_id [PK] integer	category character varying (120)	name character varying (150)	mnp numeric (8,2)	discountpercent numeric (5,2)	availablequantity integer	discountedsellingprice numeric (8,2)	weightingms integer	outofstock boolean	quantity integer
1	224	Fruits & Vegetables	Onion	2500.00	16.00	3	2100.00	1000	false	1
2	225	Fruits & Vegetables	Tomato Hybrid	4200.00	16.00	3	3500.00	1000	false	1
3	226	Fruits & Vegetables	Tender Coconut	5100.00	15.00	3	4300.00	58	false	1
4	227	Fruits & Vegetables	Coriander Leaves	2000.00	15.00	3	1700.00	100	false	100
5	228	Fruits & Vegetables	Ladies Finger	1400.00	14.00	3	1200.00	250	false	250
6	229	Fruits & Vegetables	Potato	3500.00	17.00	3	2900.00	1000	false	1
7	230	Fruits & Vegetables	Lemon	7500.00	16.00	3	6300.00	200	false	200
8	231	Fruits & Vegetables	Watermelon	5800.00	15.00	3	4900.00	58	false	1
9	232	Fruits & Vegetables	Capsicum Green	2300.00	17.00	3	1900.00	250	false	250
10	233	Fruits & Vegetables	Chilli Green	1900.00	15.00	3	1600.00	100	false	100

Null values

```
SELECT * FROM zepto
WHERE name IS NULL

OR

category IS NULL

OR

mrp IS NULL

OR

discountPercent IS NULL

OR

discountedSellingPrice IS NULL

OR

weightInGms IS NULL

OR

availableQuantity IS NULL

OR

outOfStock IS NULL

OR

quantity IS NULL;
```

Data Output Messages Notifications										
sku_id	category	name	mrp	discountpercent	availablequantity	discountedsellingprice	weightingms	outofstock	quantity	
[PK] integer	character varying (120)	character varying (150)	numeric (8,2)	numeric (5,2)	integer	numeric (8,2)	integer	boolean	integer	

Different product categories

```
SELECT DISTINCT category
FROM zepto
ORDER BY category;
```

Data Output		Messages	Notificati
	category		
	character varying (120)		
1	Beverages		
2	Biscuits		
3	Chocolates & Candies		
4	Cooking Essentials		
5	Dairy, Bread & Batter		
6	Fruits & Vegetables		
7	Health & Hygiene		
8	Home & Cleaning		
9	Ice Cream & Desserts		
10	Meats, Fish & Eggs		
11	Munchies		
12	Paan Corner		
13	Packaged Food		
14	Personal Care		

Products in stock vs out of stock

```
SELECT outOfStock, COUNT(sku_id)
```

```
FROM zepto
```

```
GROUP BY outOfStock;
```

Data Output		Messages	Notifi
	outofstock	count	
	boolean	bigint	
1	false	3279	
2	true	453	

Product names present multiple times

```
SELECT name, COUNT(sku_id) AS "Number of SKUs"
```

```
FROM zepto
```

```
GROUP BY name
```

```
HAVING count(sku_id) > 1
```

```
ORDER BY count(sku_id) DESC;
```

Data Output Messages Notifications		
	name character varying (150)	Number of SKUs bigint
1	Sunfeast Yippee! Pasta Treat - Sour Cream Onion	10
2	Arden Eggs White	10
3	Quaker Oats	10
4	Saffola Veggie Twist Masala Oats	10
5	Amul Delicious Fat Spread - Cholesterol Free	10
6	Mother's Recipe Tamarind Paste	10
7	Kellogg's Real Almond & Honey Corn Flakes	9
8	Amul Fresh Cream	8
9	iD Idli & Dosa Batter	7
10	Whisper Choice Regular Sanitary Pads	6
11	Closeup Everfresh+ Anti-Germ Gel Toothpaste Red Hot	6
12	Zorabian Chicken Kheema Parathas	6
13	Pantal Green Curry Paste	6

Data cleaning

Products with price = 0

```
SELECT * FROM zepto
```

```
WHERE mrp = 0 OR discountedSellingPrice = 0;
```

```
DELETE FROM zepto
```

```
WHERE mrp = 0;
```

Data Output	Messages	Notifications
DELETE 1		
Query returned successfully in 71 msec.		

Convert paise to rupees

```
UPDATE zepto
```

```
SET mrp = mrp / 100.0,
```

```
discountedSellingPrice = discountedSellingPrice / 100.0;
```

```
SELECT mrp, discountedSellingPrice FROM zepto;
```

Data Output Messages Notifications			
+ SQL			
	mrp numeric (8,2)	discountedsellingprice numeric (8,2)	
1	25.00	21.00	
2	42.00	35.00	
3	30.00	29.00	
4	50.00	44.00	
5	60.00	60.00	
6	425.00	383.00	
7	91.00	82.00	
8	97.00	97.00	
9	48.00	47.00	
10	225.00	210.00	
11	115.00	115.00	
12	250.00	225.00	
13	99.00	99.00	

Data analysis

-- Q1. Find the top 10 best-value products based on the discount percentage.

SELECT DISTINCT name, mrp, discountPercent

FROM zepto

ORDER BY discountPercent DESC

LIMIT 10;

Data Output Messages Notifications			
+ SQL			
	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dukes Waffy Strawberry Wafers	45.00	51.00
2	Dukes Waffy Orange Wafers	45.00	51.00
3	Dukes Waffy Chocolate Wafers	45.00	51.00
4	Epigamia Fruit Yogurt Strawberry	40.00	50.00
5	Epigamia Fruit Yogurt Vanilla	40.00	50.00
6	Chef's Basket Durum Wheat Penne Pasta	160.00	50.00
7	RRO Fresh Ricotta	275.00	50.00
8	RRO Mozzarella Block Cheese	295.00	50.00
9	Epigamia Fruit Yogurt Alphonso Mango	40.00	50.00
10	Moi Soi Sichuan Chilli Oil- For Stir Fry Marinade Spread & ...	280.00	50.00

--Q2.What are the Products with High MRP but Out of Stock

```
SELECT DISTINCT name, mrp
FROM zepto
WHERE outOfStock = TRUE and mrp > 300
ORDER BY mrp DESC;
```

Data Output Messages Notifications		
		
	name character varying (150)	mrp numeric (8,2)
1	Patanjali Cow's Ghee	565.00
2	MamyPoko Pants Standard Diapers, Extra Large (12 - 17 ...	399.00
3	Aashirvaad Atta With Mutigrains	315.00
4	Everest Kashmiri Lal Chilli Powder	310.00

--Q3.Calculate Estimated Revenue for each category

```
SELECT category,
SUM(discountedSellingPrice * availableQuantity) AS total_revenue
FROM zepto
GROUP BY category
ORDER BY total_revenue;
```

Data Output Messages Notifications		
		
	category character varying (120)	total_revenue numeric
1	Fruits & Vegetables	10846.00
2	Meats, Fish & Eggs	20693.00
3	Biscuits	25007.60
4	Beverages	55051.00
5	Dairy, Bread & Batter	55051.00
6	Health & Hygiene	64180.00
7	Home & Cleaning	122661.00
8	Ice Cream & Desserts	224385.00
9	Chocolates & Candies	224385.00
10	Packaged Food	224385.00
11	Personal Care	270849.00
12	Paan Corner	270849.00
13	Munchies	337369.00

-- Q4. Find all products where MRP is greater than ₹500 and discount is less than 10%.

SELECT DISTINCT name, mrp, discountPercent

FROM zepto

WHERE mrp > 500 AND discountPercent < 10

ORDER BY mrp DESC, discountPercent DESC;

Data Output Messages Notifications			
	name character varying (150)	mrp numeric (8,2)	discountpercent numeric (5,2)
1	Dhara Kachi Ghani Mustard Oil Jar	1250.00	8.00
2	Saffola Gold (Jar)	1240.00	0.00
3	Dhara Filtered Groundnut Oil (Jar)	1050.00	1.00
4	Fortune Rice Bran Health Oil (Jar)	1050.00	1.00
5	Dhara Filtered Groundnut Oil (Jar)	1050.00	0.00
6	Fortune Soyabean Oil	1005.00	0.00
7	Fortune Sunlite Refined Sunflower (Jar)	925.00	0.00
8	Surf Excel Matic Powder Front Load	810.00	7.00
9	Surf Excel Matic Top Load	720.00	9.00
10	Pedigree Puppy Dry Dog Food Food Chicken & Milk	690.00	6.00
11	Pedigree Dog Food Adult Meat & Rice	660.00	7.00
12	Lizol Double Concentrate Disinfectant Floor Cleaner Citrus	650.00	8.00
13	Nestle Nestle Nan Pro Follow-Up Formula	650.00	0.00

-- Q5. Identify the top 5 categories offering the highest average discount percentage.

SELECT category,

ROUND(AVG(discountPercent),2) AS avg_discount

FROM zepto

GROUP BY category

ORDER BY avg_discount DESC

LIMIT 5;

Data Output Messages Notifications		
	category character varying (120)	avg_discount numeric
1	Fruits & Vegetables	15.46
2	Meats, Fish & Eggs	11.03
3	Ice Cream & Desserts	8.32
4	Chocolates & Candies	8.32
5	Packaged Food	8.32

-- Q6. Find the price per gram for products above 100g and sort by best value.

```
SELECT DISTINCT name, weightInGms, discountedSellingPrice,
ROUND(discountedSellingPrice/weightInGms,2) AS price_per_gram
FROM zepto
WHERE weightInGms >= 100
ORDER BY price_per_gram;
```

Data Output Messages Notifications				
Showing rows: 1 to 1000				
	name character varying (150)	weightInGms integer	discountedSellingPrice numeric (8,2)	price_per_gram numeric
1	Tata Salt	1000	24.00	0.02
2	Onion	1000	21.00	0.02
3	Onion	3000	57.00	0.02
4	Shubh kart - Nirmal sugandhi mogra wet dhoop zipper 20 sticks	1160	28.00	0.02
5	Vicks Cough Drops Menthol	1160	20.00	0.02
6	Aashirvaad Iodised Salt	1000	19.00	0.02
7	Carrot	500	15.00	0.03
8	Potato	1000	29.00	0.03
9	Beetroot	500	13.00	0.03
10	Baby Potato	500	16.00	0.03
11	Shubh kart - Tejas Twisted Cotton Wicks 1000n	1000	28.00	0.03
12	Raw Banana	500	17.00	0.03
13	Potato	3000	84.00	0.03

--Q7.Group the products into categories like Low, Medium, Bulk.

```
SELECT DISTINCT name, weightInGms,
CASE WHEN weightInGms < 1000 THEN 'Low'
WHEN weightInGms < 5000 THEN 'Medium'
ELSE 'Bulk'
END AS weight_category
FROM zepto;
```

Data Output Messages Notifications				
Showing rows: 1 to 1000				
	name character varying (150)	weightInGms integer	weight_category text	
1	Popular Essentials Poppy Seeds (Gasagase)	100	Low	
2	Origami Kitchen Towels 2 In 1 Quilted 2ply	116	Low	
3	Origami Good Karma Paper Serviettes - 100 napkins (1 ply)	58	Low	
4	Johnson's Baby Top To Toe Bath Body Wash	210	Low	
5	Harpic Hygienic Toilet Cleaner Rim Block Citrus	26	Low	
6	Lizol Disinfectant Surface & Floor Cleaner Liquid Jasmine	500	Low	
7	Sapota	348	Low	
8	Britannia Treart Creme Wafers Chocolate	60	Low	
9	Popular Essentials Dhanla (Coriander Seeds)	200	Low	
10	Popular Essentials Californian Almond	250	Low	
11	Everest Shahi Biryani Masala	50	Low	
12	Keya Italian Pizza Oregano	80	Low	
13	Figaro Plain Green Olive	450	Low	

Total rows: 1782 Query complete 00:00:00.202

--Q8.What is the Total Inventory Weight Per Category

```
SELECT category,  
SUM(weightInGms * availableQuantity) AS total_weight  
FROM zepto  
GROUP BY category  
ORDER BY total_weight;
```

Data Output Messages Notifications		
	category character varying (120)	total_weight bigint
1	Meats, Fish & Eggs	48016
2	Biscuits	84431
3	Fruits & Vegetables	91794
4	Health & Hygiene	142904
5	Dairy, Bread & Batter	143735
6	Beverages	143735
7	Paan Corner	348187
8	Personal Care	348187
9	Home & Cleaning	373161
10	Ice Cream & Desserts	490797
11	Chocolates & Candies	490797
12	Packaged Food	490797

--Q9.Top 10 products generating highest estimated revenue

```
SELECT name,  
category,  
SUM(discountedSellingPrice * availableQuantity) AS estimated_revenue  
FROM zepto  
GROUP BY name, category  
ORDER BY estimated_revenue DESC  
LIMIT 10;
```

Data Output Messages Notifications			
	name character varying (150)	category character varying (120)	estimated_revenue numeric
1	Borges Extra Light Olive Oil Bottle	Cooking Essentials	8394.00
2	Borges Extra Light Olive Oil Bottle	Munchies	8394.00
3	Dhara Filtered Groundnut Oil (Jar)	Munchies	8310.00
4	Dhara Filtered Groundnut Oil (Jar)	Cooking Essentials	8310.00
5	Praakritik Natural Desi Gir Cow A2 G...	Cooking Essentials	7830.00
6	Praakritik Natural Desi Gir Cow A2 G...	Munchies	7830.00
7	Saffola Gold (Jar)	Munchies	7440.00
8	Saffola Gold (Jar)	Cooking Essentials	7440.00
9	Dhara Kachi Ghani Mustard Oil Jar	Munchies	6858.00
10	Dhara Kachi Ghani Mustard Oil Jar	Cooking Essentials	6858.00

--Q10.Categories with the most out-of-stock products

SELECT category,

COUNT(*) AS out_of_stock_products

FROM zepto

WHERE outOfStock = TRUE

GROUP BY category

ORDER BY out_of_stock_products DESC;

Data Output Messages Notifications		
	category character varying (120)	out_of_stock_products bigint
1	Munchies	64
2	Cooking Essentials	64
3	Packaged Food	45
4	Ice Cream & Desserts	45
5	Chocolates & Candies	45
6	Biscuits	42
7	Beverages	28
8	Dairy, Bread & Batter	28
9	Personal Care	21
10	Paan Corner	21
11	Home & Cleaning	19
12	Health & Hygiene	13
13	Meats, Fish & Eggs	12